Higher National Diploma in Engineering (Civil)

Third Year, Second Semester Examination - 2022

CE 3117-Concrete Technology

(01.)

I. Briefly explain about the Segregation & Bleeding in fresh concrete.

II. Mention 03 characteristics of a good formwork. (03 Marks)

III. "Use of Glass reinforced plastic formwork is a new trend in the construction industry". Justify the statement by highlighting its advantages over the alternative formwork materials. (05 Marks)

IV. Briefly discuss about the "Prestressed concrete technique". (04 Marks)

V. State the advantages and disadvantages in Pre-tensioning process. (02 x 2 Marks)

(02.)

I. Briefly explain about the following Cement types and their applications in the industry. (02 x 4 Marks)

a) Ordinary Portland cement

b) Portland Pozzolana cement

c) Rapid Hardening Cement

d) Low Heat Cement

II. "At present cement supply is not sufficient to cater the demand in construction industry. Hence, contractors try to collect required cement quantity without considering cement type or its brand." Discuss the effects of above practice in the construction industry. (04 Marks)

III. Briefly describe the Cement manufacturing process. (Clearly mention the raw materials, used in manufacturing process) (03 Marks)

IV. State the effects of various constituents of cement. (05 Marks)

(03.)

I. Name 02 methods of cement deterioration. (02 Marks)

II. Discuss the effects of following characteristics of aggregate relevant to the better concrete mixture preparation. (02 x 4 Marks)

a) Durability

b) Cleanliness

c) Sizes of aggregates d) Grading of aggregates

III. State 04 types of tests which determine the quality of aggregates. (04 Marks)

IV. Briefly explain the function and applications of the following concrete admixtures. (02 x 3 Marks)

a) Retarding admixtures

b) Silica Fume

c) Fly ash

(04.)

I. Define "Concrete deterioration" in concrete structures. (02 Marks)

II. Discuss about the following Concrete deterioration methods. (Clearly mention the requirements for reactions, mechanism and adverse effects) (03 x 4 Marks)

a) Alkali-silica reaction

b) Sulphate attack

c) Delayed ettringite formation

d) Chloride Ingress

III. State the 04 methods of avoiding corrosion in the reinforced concrete. (04 Marks)

IV. Mention 04 methods of increasing the fire resistance of concrete structures. (02 Marks)

(05.)

I. How does the Concrete creep influences on the structural elements? (04 Marks) 43

II. Define and explain "Plastic shrinkage effect" in concrete. (04 Marks) 44

III. Mention 04 methods of preventing the cause of plastic shrinkage. (04 Marks) 45

IV. Briefly describe the following mechanical properties of concrete. (02 x 4 Marks)

a) Compressive strength

b) Flexural strength

c) Shear strength

d) Modulus of elasticity

(06.)

I. Mention 03 qualities in a properly proportioned concrete mix. (03 Marks)

II. Calculate the quantities for a trial mix which appropriates to a concrete mixture Grade 30, having workability of 30-60 mm in terms of slump test. Used cement type is O.P.C. (strength class 42.5). Maximum size of the crushed coarse aggregate is 20 mm.In addition, river sand was used as fine aggregate with estimated percentage of passing 600μm sieve as 50% from the total aggregate content.Relative density for combined aggregate density is 2.65. Assume 2.5% defective and Standard Deviation as 4.0. Maximum W/C ratio is 0.5 & minimum cement content i350kg/m^3 as per the specification of BS 8110: Part I.

You may use following tables and graphs for calculations.

Clearly mention all the references and assumptions. (14 Marks)

III. Discuss the possible reasons for mis-matching the theoretical data and practical outputs used in the mix design. (03 Marks)

Table 01: Probability factor

Defectives %	K value
10	1.28 [61]
5	1.64 [62]
2.5	**1.96** [63]
1	2.33 [64]

Table 02: Compressive strength of concrete mixes made with a free-water cement ratio of 0.5

Cement strength class	*Type of coarse aggregate*	*Compressive strength (N/mm²)*			
		Age (days)			
		3	*7*	*28*	*91*
42.5	Uncrushed	22	30	42	49
	Crushed	27	36	49	56
52.5	Uncrushed	29	37	48	54
	Crushed	34	43	55	61

Table 3: Approximate free water content required to give various levels of workability

Maximum size of aggregate (mm) (in)	Slump (mm)	0-10 > 12	10-30	30-60	60-180
	V-B (sec)		6-12	3-6	0-3
	Type of aggregates				
10(3/8)	Uncrushed	150	180	205	225
	Crushed	180	205	230	250
20(3/4)	Uncrushed	135	160	180	195
	Crushed	170	190	210	225
40(3/2)	Uncrushed	115	140	160	175
	Crushed	155	175	190	205

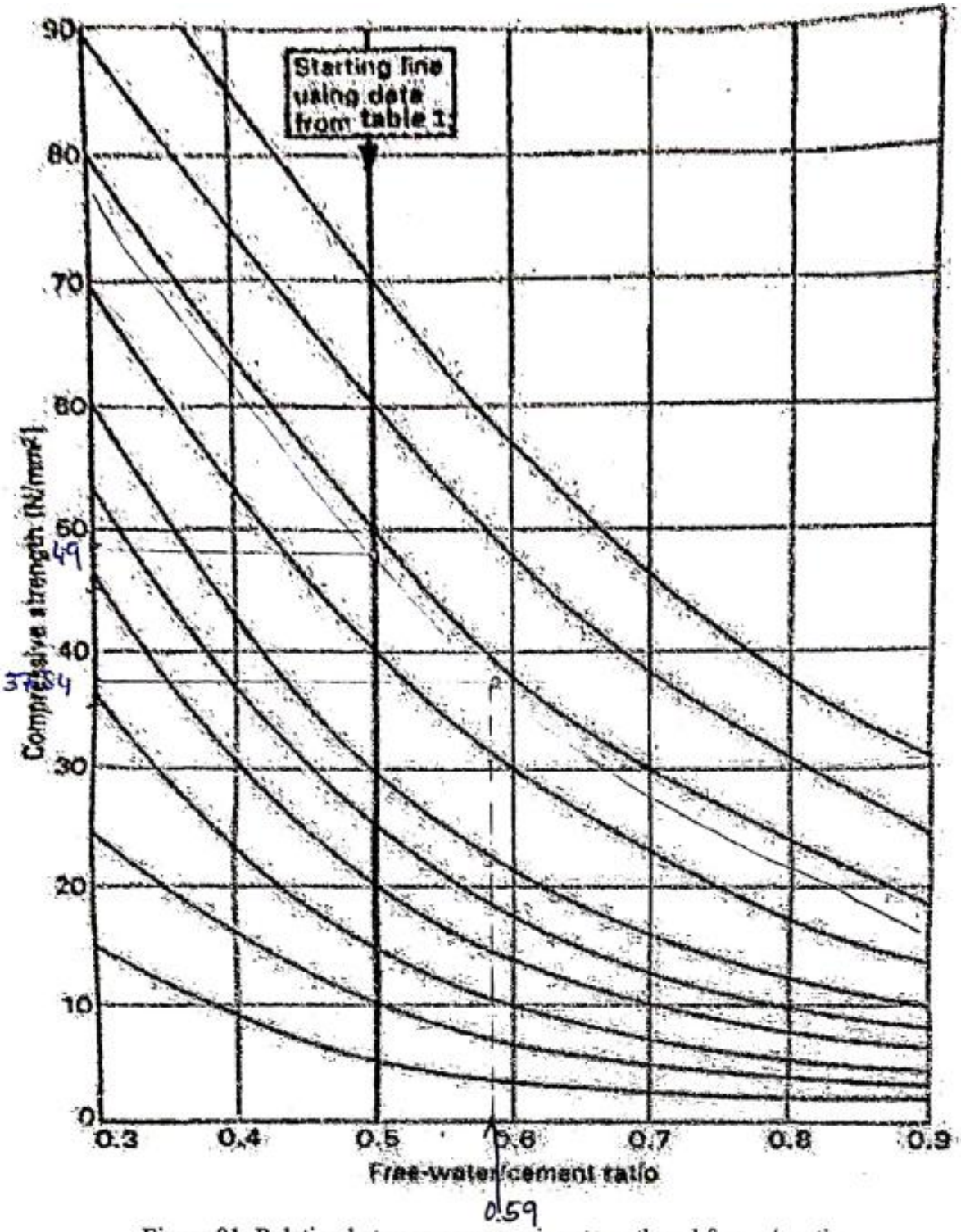

Figure 01: Relation between compressive strength and free w/c ratio

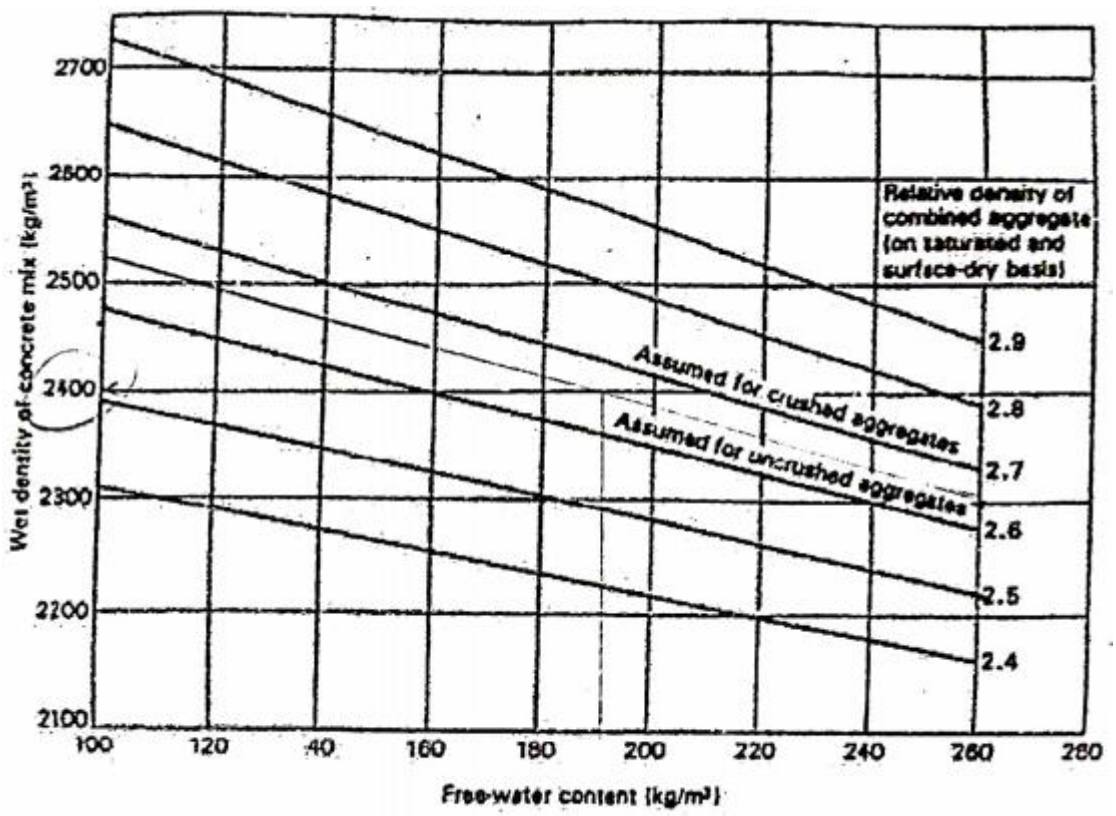

Figure 02: Estimated wet density for fully compacted concrete

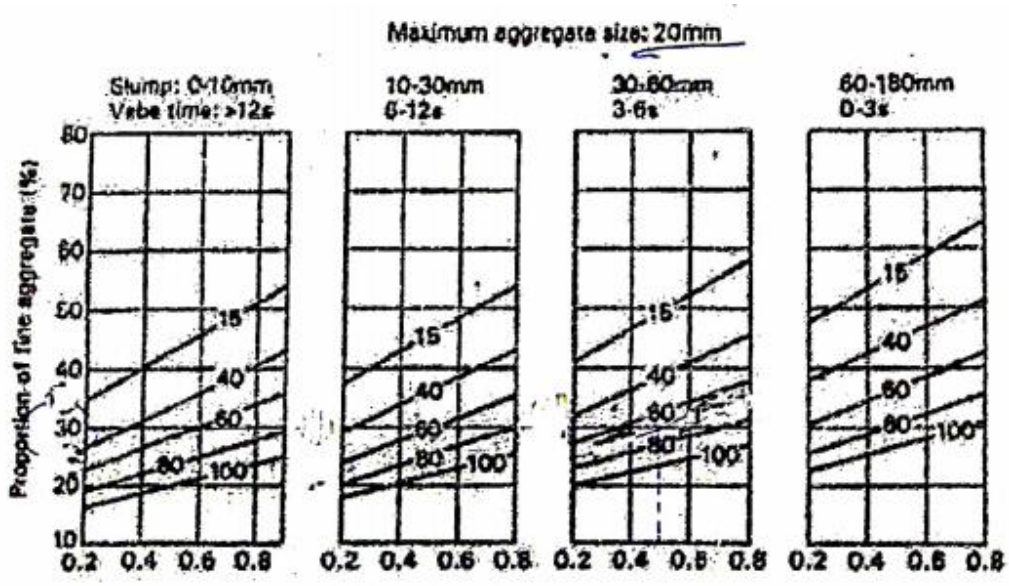

Figure 03: Recommended % of fine aggregate as a function of free w/c ratio for various values of workability and mix